When undertaking control of locusts bands and swarms, the APLC applies very low doses of insecticide that degrade rapidly in the environment. To safeguard against the risk of excessive insecticide residues in grain or stock in areas where locust control agents have been applied, strict withholding periods or slaughter intervals are imposed. This allows ample time for the insecticides to break down to below the levels set for each pesticide used.

Fact sheets are available below for the insecticides currently used by the APLC for aerial control. They provide information on the use, label restrictions and effects of the formulations of these insecticides.

[expand all]

Fenitrothion

Information on the insecticide Fenitrothion (Sumithion ULV), Australian Plague Locust Commission (APLC) May 2008

The Australian Plague Locust Commission uses fenitrothion (tradename - Sumithion ULV) applied by aircraft, for control of locust bands and swarms. The formulation used is suitable only for ultra-low volume (ULV) application, and cannot be mixed with water for conventional ground spraying.

Sumithion ULV contains 1.27 kg of fenithrothion (the active ingredient)/L.

The label permits the application of fenitrothion at rates of 200-400 mL (254-508 g a.i.) /ha. However, APLC research has found that a lower rate of 210-300 mL (267-381 g a.i.) /ha is effective and this is used operationally.

Fenitrothion is registered for use against locusts on pasture and a wide range of cereal and other crops eg. grazing sorghum, lucerne, apples, grapes, lettuce, tomatoes and soybean.

The registered label conditions impose a 1.5 km no spray zone between a sprayed block and any sensitive area downwind, eg dwellings, dams, rivers, organic production.

Fenitrothion is an organophosphorus insecticide, of moderate mammalian toxicity and acts on the nervous system of the insect. Fenitrothion is degraded rapidly in livestock, within 2-7 days according to Australian and overseas research. On vegetation the level of any residue is halved every day. However, re-entry into treated areas during the 48 hours immediately following application should be restricted. APLC Officers re-entering treated areas during this period to assess efficacy must take precautions and wear suitable protective clothing.

Research has demonstrated that fenitrothion has no major or long-term impacts on invertebrates of grasslands.

Fenitrothion is highly toxic to aquatic invertebrates (e.g. shrimps, insects & yabbies) and honey bees. It is of moderate toxicity to fish and birds.

Following the application of fenitrothion to crop or pasture the following withholding periods and/or slaughter interval must be observed in accordance with the current (2000) registered label:

- for pasture & lucerne, do not graze for 7 days after application.
- do not cut for stockfeed for 14 days after application.
- do not apply later than 14 days before harvest (edible crops).
- locusts may be treated in an area where stock are grazing (oversprayed) as long as the animals are not be sent for slaughter until 14 days after the pasture was sprayed.

Fenitrothion (both ULV and emulsifiable concentrate [EC] formulations) can cause leaf discoloration and burning to some sorghum varieties.

ALWAYS CHECK AND ADHERE TO THE REQUIREMENTS OF THE CURRENT REGISTERED LABEL OR OFF-LABEL PERMIT FOR ALL AGRICULTURAL CHEMICALS.

Download

Document	Pages	File size
Fenitrothion fact sheet PDF	1	27 KB

If you have difficulty accessing these files, visit web accessibility for assistance.

Fipronil

Information on the insecticide Fipronil, Australian Plague Locust Commission (APLC) May 2008

Fipronil (tradenames, Adonis 3UL and Regent 200SC) is an insecticide for control of locusts and grasshoppers. The APLC began using Adonis 3UL (the ULV formulation suitable for aerial application) operationally in 2000. It has proven to be highly effective when applied by aircraft using a wide interval application technique (spray runs spaced at intervals of 200 to 500 m).

Fipronil is a phenyl pyrazole, a new class of pesticide that acts on the insect nervous system. It kills by contact or ingestion. Fipronil is highly active against locusts and grasshoppers which means that very low doses (0.3-1 g a.i./ha) can be used. By comparison, organophosphorous pesticides used for locust control are applied in the range of 250-500 g a.i./ha.

Fipronil is moderately toxic to mammals and in its technical form is slightly more toxic than the organophosphorous pesticide fenitrothion. However, the low doses of fipronil used by the APLC minimise the risk to terrestrial vertebrates. The ULV formulation contains 3 g a.i./L and the suspension concentrate 200 g a.i./L compared with at least 1,000 g a.i./L for organophosphorous pesticides.

Fipronil is highly toxic to bees, termites and aquatic invertebrates such as yabbies. While it is unlikely to affect fish at the registered dose, it must not be applied near waterways. The APLC will not target areas where significant numbers of termite mounds are evident. The APLC also uses a 1.5 km no spray zone between a target treated by aerial spraying and any sensitive areas downwind, eg. dwellings, dams, waterways, organic production.

Fipronil is more persistent than the organophosphorous pesticides. It will kill locusts invading sprayed areas for up to a week after application but this also means that grazing stock may be exposed to it for longer periods. After consumption by cattle it remains in the fat longer than the organophosphorous compounds, breaking down with a half-life of about 7 days. Residues in cattle are not a concern if fipronil is applied correctly and withholding periods from grazing strictly abided by.

Fipronil has been registered by the Australian Pesticides and Veterinary Medicines Authority (APVMA) as Adonis 3UL (ULV formulation) & Regent 200 SC (water miscible formulation) and can only be used to treat locusts feeding on pasture or sorghum (forage and grain). It is not registered for use on cereal crops.

The current (2006) registered label lists the following withholding periods and/or slaughter interval:

- Pasture: Do not graze or cut for stock food for 14 days after application.
- Sorghum (forage & grain): Do not harvest, graze or cut for stock food for 14 days after application.
- Livestock: Withhold stock from slaughter for 21 days after application, where stock were present in crop or pasture at time of application.

ALWAYS CHECK AND ADHERE TO THE REQUIREMENTS OF THE REGISTERED LABEL OR OFF-LABEL PERMIT FOR ALL AGRICULTURAL CHEMICALS.

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Document	Pages	File size
Fipronil fact sheet PDF	1	36 KB

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Metarhizium

Information on the Bio-insecticide Green Guard® ULV, Australian Plague Locust Commission (APLC) August 2010

Green Guard® ULV is a bio-insecticide containing a naturally occurring Australian fungus, Metarhizium anisopliae var. acridum, mixed with a spray oil. It is produced by Becker Underwood Pty.

Ltd. for the biological control of locusts and grasshoppers and is registered for this use by the Australian Pesticides and Veterinary Medicines Authority (APVMA).

The APLC began using Green Guard® ULV operationally in 2000 and since then it has been used extensively where there are land use limitations on chemical pesticides such as near waterways, wetlands or certified organic production properties.

Green Guard® ULV is produced as a concentrate of fungal spores suspended in corn oil. It is mixed with Summer Spray Oil and applied by aircraft using ULV (ultra low volume) spraying equipment. The Green Guard®& Summer Spray Oil blend has been passed as suitable for use on organic properties by the National Association for Sustainable Agriculture-Australia (NASAA).

The standard dose for control of Australian plague locust is 25g of spores in 700 mL of oil per hectare. In cooler conditions, tall or thick vegetation, a dose of 30 to 75 g/ha may be required.

Locusts and grasshoppers are infected when Metarhizium spores attach to the insect. The spores then germinate and penetrate the cuticle growing into the body and internal organs of the insect. Mortality usually occurs 8-21 days after treatment, though this can take longer when temperatures are cool. The delay in mortality means it is advisable to treat when locusts and grasshoppers are in the early nymphal stages and less mobile than the winged adults.

The Metarhizium isolate present in Green Guard® is relatively specific to locusts and grasshoppers. However, there is a potential hazard to honeybees as they may collect high numbers of spores while foraging on flowering plants. To limit this possibility, aerial application of Green Guard® by the APLC will not occur within 5,000 m of active beehives. .

No withholding periods and/or slaughter intervals apply when using Green Guard®.

As a precaution to limit the possibility of unintended drift from aerial spraying near waterways and other sensitive areas, the current (2004) registered label specifies use of a 100 m no-spray zone between a treated area and any wetland, water body or watercourse downwind.

ALWAYS CHECK and ADHERE TO THE REQUIREMENTS OF THE CURRENT REGISTERED LABEL OR OFF-LABEL PERMIT FOR ALL AGRICULTURAL CHEMICALS.

Download

Document	Pages	File size
Metarhizium fact sheet PDF	1	30 KB

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Source: https://www.agriculture.gov.au/biosecurity-trade/pests-diseases-weeds/locusts/landholders/control-agents-residues